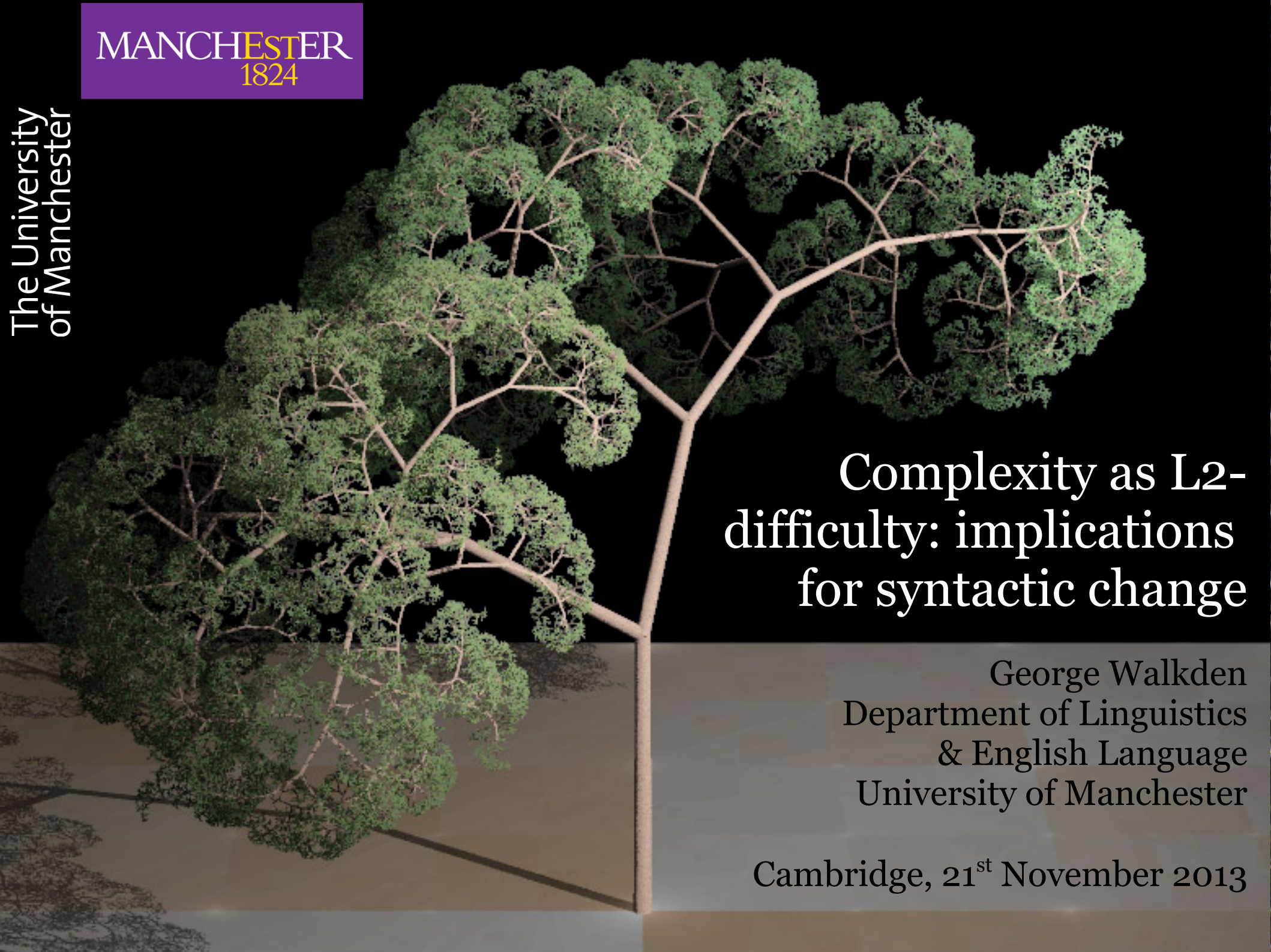




Complexity as L2- difficulty: implications for syntactic change

George Walkden
Department of Linguistics
& English Language
University of Manchester

Cambridge, 21st November 2013

Talk outline

- Research question: how does the “Trudgill insight” about language contact and change relate to syntax?
 - Evidence from three domains:
 - Negation/Jespersen’s Cycle
 - Null subjects
 - Case (very briefly!)

Introduction

Negation

Null subjects

Case

Conclusion

A puzzle

- What are the structural effects of language contact on the languages that emerge from it?

- Sociolinguists:

contact leads to
simplification

- Typologists:

contact leads to
complexification

Introduction

Negation

Null subjects

Case

Conclusion

The equicomplexity hypothesis

- Both views entail rejecting the **equicomplexity hypothesis**, the view that

“the total grammatical complexity of any language, counting both morphology and syntax, is about the same as any other”

(Hockett 1958: 180–1)

Trudgill's solution

- The 'Trudgill insight': **different types of language contact situation may give rise to different types of change** (Trudgill 2011).
- In particular:
 - short-term adult (L2) language contact tends to lead to **simplification**
 - long-term, co-territorial language contact tends to lead to **additive complexification**
 - isolation tends to lead to **spontaneous complexification**

Examples of evidence

- Short-term L2 contact > **simplification**:
 - loss of morphological distinctions in verb forms in Nubi Creole Arabic, spoken in Kenya & Uganda (Owens 1997, 2001)
- **complexification**:
 - **Additive**: development of case marking and 5-way evidential system in Amazonian language Tariana under the influence of Tucano (Aikhenvald 2003)
 - **Spontaneous**: exaptation of –y ending for intransitive infinitives in dialects of the south-west of England (Ihalainen 1991)

Syntax

- Trudgill (2011) deploys case studies from morphology and (in chapter 5) phonology
- Syntax is not considered (nor is semantics)
- Aim of today's talk:
 - If correct, does Trudgill's insight carry over into historical syntax?
 - If so, how can this be understood in terms of syntactic theory?

Logical possibilities

- a) The Trudgill insight is wrong.
- b) The Trudgill insight is correct for some domains of language (e.g. morphology), but not others (e.g. syntax)
- c) The Trudgill insight is correct for syntax, but not in a way that syntactic theory helps us capture.
- d) The Trudgill insight is correct for syntax, and has a neat theoretical characterization.

Potential objections, 1

- “Our theory doesn’t allow us to talk about differences in complexity.”
 - cf. Trotzke & Zwart (2013): “Within the generative paradigm, the question of the comparative complexity of languages, as discussed in recent typological-functional literature, does not arise”
 - Reasoning: complexity is determined by innate biological mechanisms, which are uniform.
 - Fallacy: this does not mean that **acquired** components of language cannot vary in complexity.

Potential objections, 1

- “Our theory doesn’t allow us to talk about differences in complexity.”
 - In a classical Principles-&-Parameters model, with a finite universal set of binary equipollent parameters, this might be true.
 - But Minimalism has moved towards a model in which all variation is lexical (the ‘**Borer-Chomsky Conjecture**’, Borer 1984, cf. Baker 2008: 353).
 - Why should the complexity of the lexicons of different speakers/languages be uniform?

Potential objections, 2

- “The notion of complexity is meaningless.”
 - The notion certainly needs to be given some content. Different kinds of complexity in syntax:
 - **Representational** complexity (Roberts & Roussou 2003: 201; van Gelderen 2009, 2011)
 - **Derivational** complexity (Mobbs 2008)
 - **Node counts** (Szmrecsanyi 2004)
 - **Processing complexity** (Hawkins 1994, 2004)
 - Strong/weak **generative capacity** (Chomsky 1963)

Potential objections, 3

- “The question is not relevant to I-language.”
 - True; all diachronic linguistics necessarily falls outside the domain of I-language.
 - But lessons learned from the study of I-language ought to be taken into account when studying diachronic issues.
 - “Internalist biolinguistic inquiry does not, of course, question the legitimacy of other approaches to language, any more than internalist inquiry into bee communication invalidates the study of how the relevant internal organization of bees enters into their social structure. The investigations do not conflict; they are mutually supportive.” (Chomsky 2001: 41)

Potential objections, 4

- “There already exist good internal explanations for these changes, without invoking language contact.”
 - The dogma that language contact should only be invoked as a last resort is open to criticism (see e.g. Farrar & Jones 2002).
 - In any case, as pointed out by Hale (1998), *all* internal accounts of language change fail to account for why the change would take place at a particular time, in a particular place.
 - The present approach can narrow this down much more.

Complexity as L2-difficulty

- Without denying the relevance of other types or definitions of complexity to language change, we're here going to define complexity as **L2-difficulty** (following Trudgill 2011, Dahl 2004).
 - L2-difficulty is simply how difficult a natural language (or aspect of language) is to acquire for post-critical-period learners.
 - Has its roots in early biolinguistics (Lenneberg 1967): L1 and L2 acquisition appear to be qualitatively (or at least quantitatively) different from one another.
 - L2-difficulty = greater **complexity**;
L2-ease = greater **simplicity**.

What counts as L2-difficult?

- A number of hypotheses exist in the generative psycholinguistic literature:
 - **Full Transfer/Full Access/Full Parse** (Schwartz & Sprouse 1994, 1996)
 - **Bottleneck Hypothesis** (Slabakova 2006, 2008, 2009)
 - **Interface Hypothesis** (Sorace 2000, 2003, 2011)
 - **Feature Reassembly Hypothesis** (Lardiere 2005, 2008, 2009)
 - **Interpretability Hypothesis** (Hawkins & Hattori 2006; Tsimpli & Dimitrakopoulou 2007)

L2-difficulty irrelevant to syntax?

- Full Transfer/Full Access/Full Parse:
 - Initial state of L2 is constrained only by final state of L1.
 - In case of parsing failure, all universally permitted options are available.
 - If this model is correct, L2-difficulty is always relativized to individual L1s and so **no broad generalizations are possible about the outcomes of contact**.
 - Bottleneck and Interface Hypotheses make similar (non-)predictions.
 - As a result, it's more interesting to explore alternatives.

The Interpretability Hypothesis

- **Uninterpretable** features not selected during L1 acquisition are no longer available in post-critical period L2 acquisition.
- Support:
 - L1 Japanese speakers learning English at an advanced level fail to acquire [**u**Wh] feature on C⁰ (Hawkins & Hattori 2006)
 - L1 Greek speakers learning English at an advanced level do not disprefer resumptive pronouns in wh-constructions at a target-like level (Tsimpili & Dimitrakopoulou 2007)
 - L1 Arabic speakers learning English at an advanced level do not accept grammatical examples of VP ellipsis (Al-Thubaiti 2009)

Arguments against the IH

- Empirical concerns:
 - L1 Spanish, Mandarin & Japanese speakers learning English at an advanced level manage to acquire English past tense functional morphology (Campos Dintrans 2011)
- Conceptual:
 - See e.g. Lardiere (2009), among many others.
- I'll be adopting a (simplified) version of the IH, hypothesizing that **uninterpretable** features are (in general!) **hard to learn** for L2 learners.

Predictions for syntactic change

- Given this (IH-like) characterization of **L2-difficulty**, then in situations involving high levels of adult L2 acquisition and transmission of adult L2 varieties, **at the population level, uninterpretable features will be lost.**
- These are the short-term contact situations discussed by Trudgill (2011), i.e. contact situations not involving high levels of balanced bilingualism.
- We will look at three potential case studies of this.

Case study 1

- From stage II to stage III of Jespersen's Cycle.
 - Jespersen's Cycle = a development in which
 - a single negative particle exists (Stage I)
e.g. *ne* (Old French)
 - it is then joined by another one (Stage II)
e.g. *ne* ... *pas* (Classical French)
 - the original particle is then lost (Stage III)
e.g. *pas* (Colloquial French)
 - Here I focus on the transition from stage II to stage III, leaving stage I aside.

Analysis of stage II of JC

- The **original** negator in Neg⁰ bears a [**u**Neg] feature; the '**new**' negator in SpecNegP bears an [**i**Neg] feature.
 - (see e.g. Wallage 2008 for Middle English; Breitbarth 2008, 2013a for Middle Low German; based on Zeijlstra 2004).
 - Focusing on Middle Low German:
 - Preverbal (head) negator: **en/ne** [**u**Neg]
 - Postverbal negator: **nicht** [**i**Neg]

Analysis of stage II of JC

- Example:

- *We des nicht enwete de latis sik berichten*
who this.GEN NEG NEG-know REL let-it REFL report
'Everyone who does not yet know this should endeavour to
learn about it' (Braunschweig, 1349)

Introduction

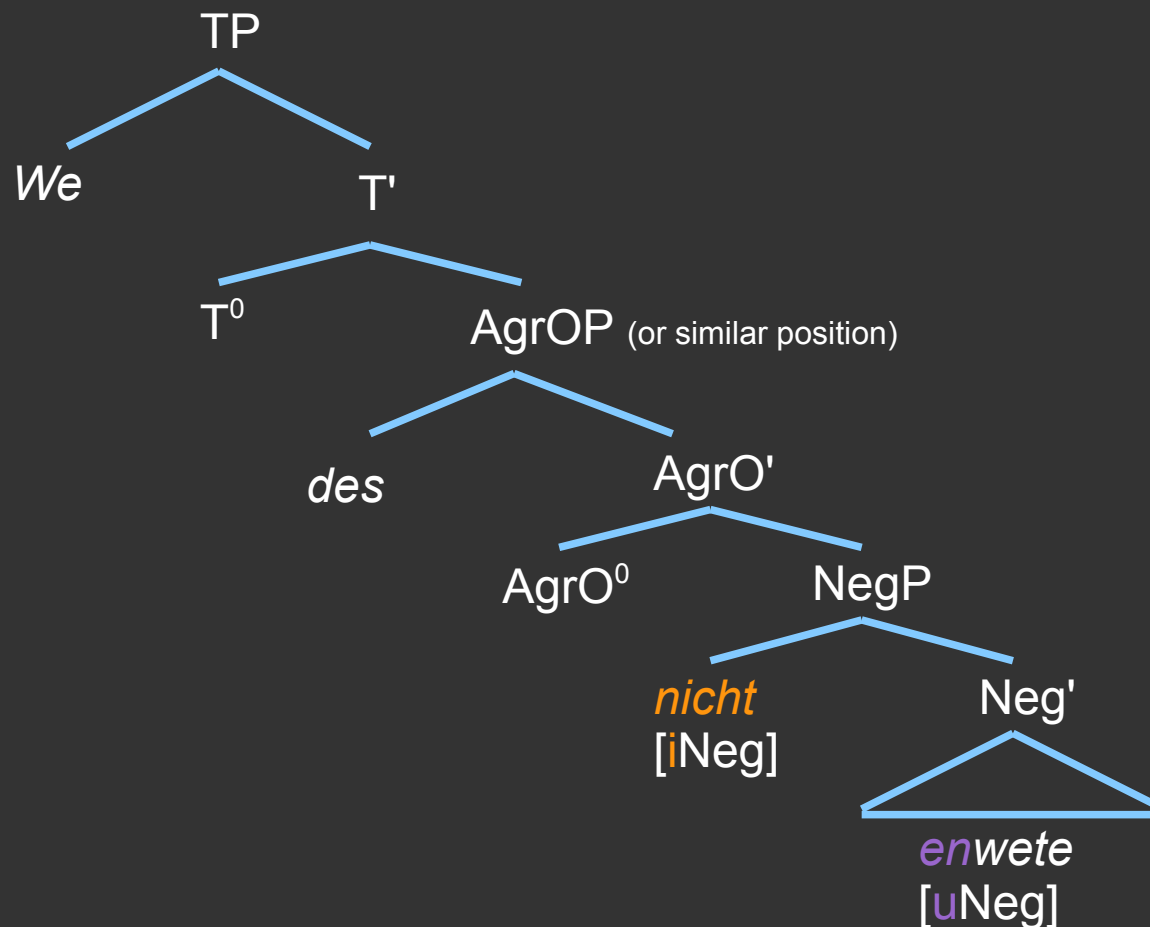
Negation

Null subjects

Case

Conclusion

Analysis of stage II of JC



Introduction

Negation

Null subjects

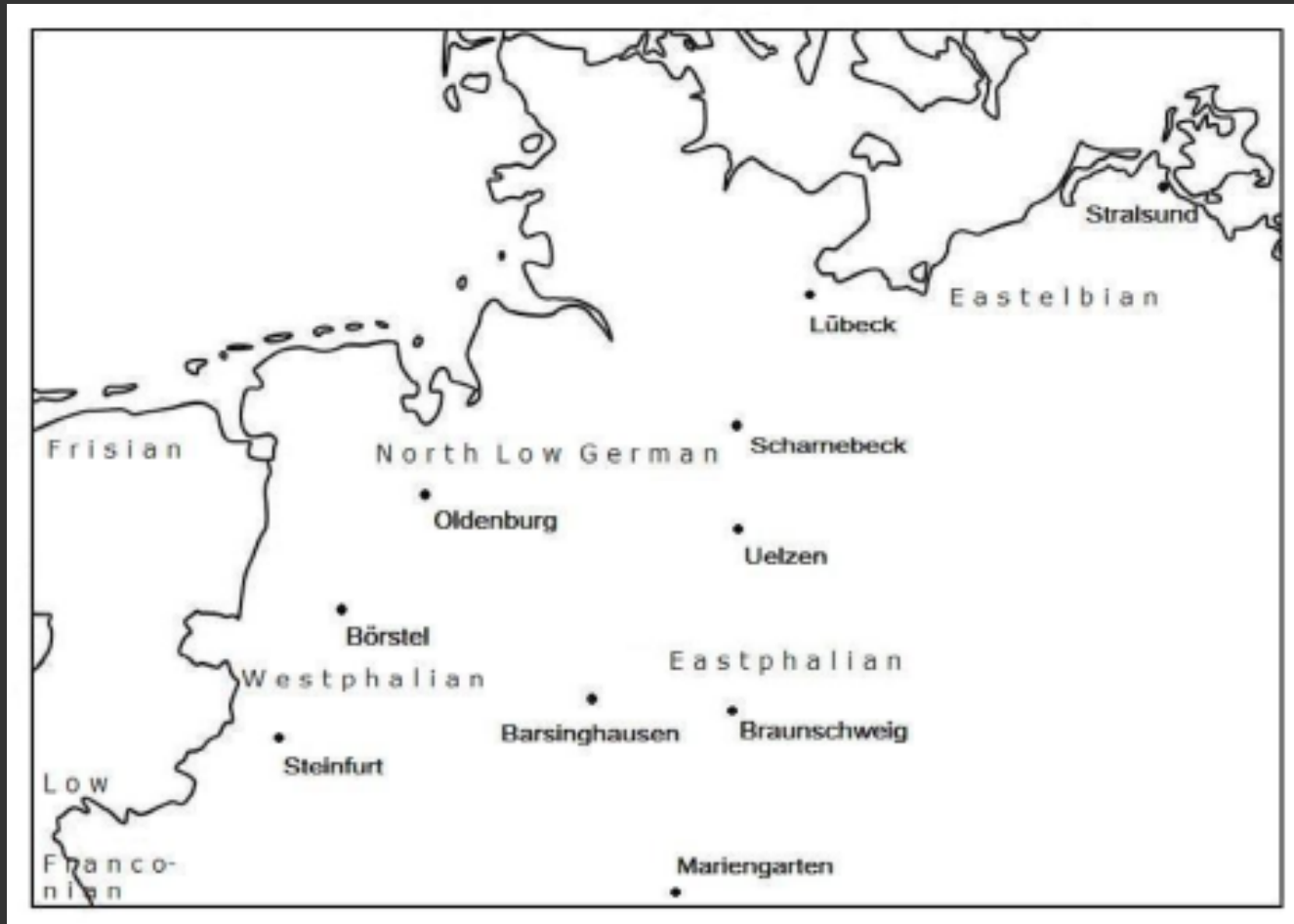
Case

Conclusion

The transition to stage III

- Stage III is also attested in Middle Low German:
 - we sek des **nicht** leddigen wel...
who REFL this.GEN NEG rid will
'Whoever does not want to rid him/herself of this...'
(Braunschweig, 1380)
- Crucially, different scribal dialects make the change at different speeds.

MLG scribal dialects



Introduction

Negation

Null subjects

Case

Conclusion

The transition to stage III

- Use of stage II by scribal dialect (Breitbarth 2013a):

	Westphalian	Eastphalian	North Low Saxon	Hansa cities
1325–1374	22 (78.6%)	56 (72.7%)	37 (56.1%)	3 (50%)
1375–1424	25 (83.3%)	52 (71.2%)	42 (33.1%)	12 (18.5%)
1425–1474	3 (37.5%)	25 (52.1%)	75 (33.0%)	20 (29%)
1475–1524	14 (35.8%)	15 (14.6%)	62 (31.2%)	10 (7.8%)
1525–1574	8 (21.1%)	18 (10.2%)	3 (12%)	2 (12.5%)

Introduction

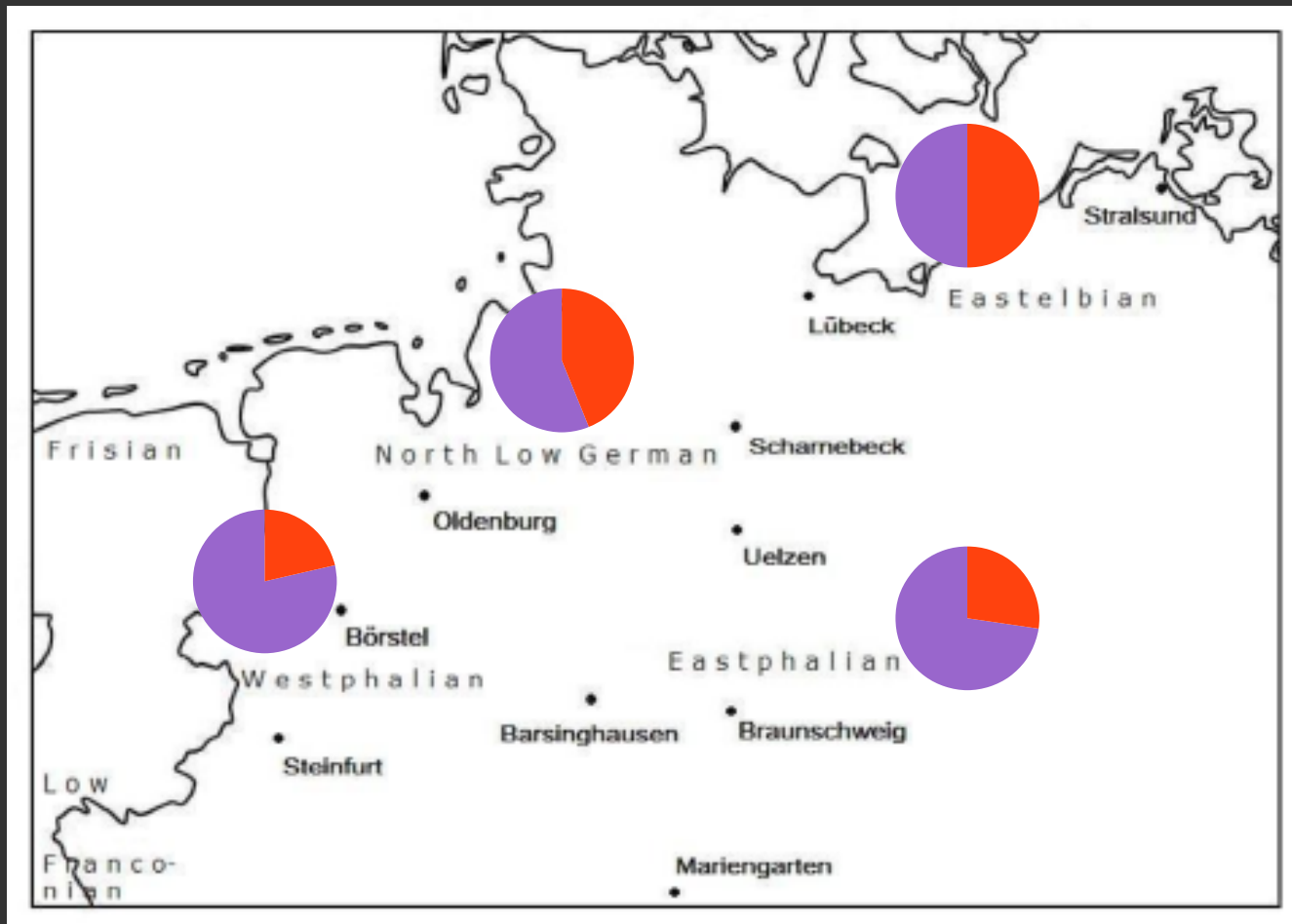
Negation

Null subjects

Case

Conclusion

1325–1374



Introduction

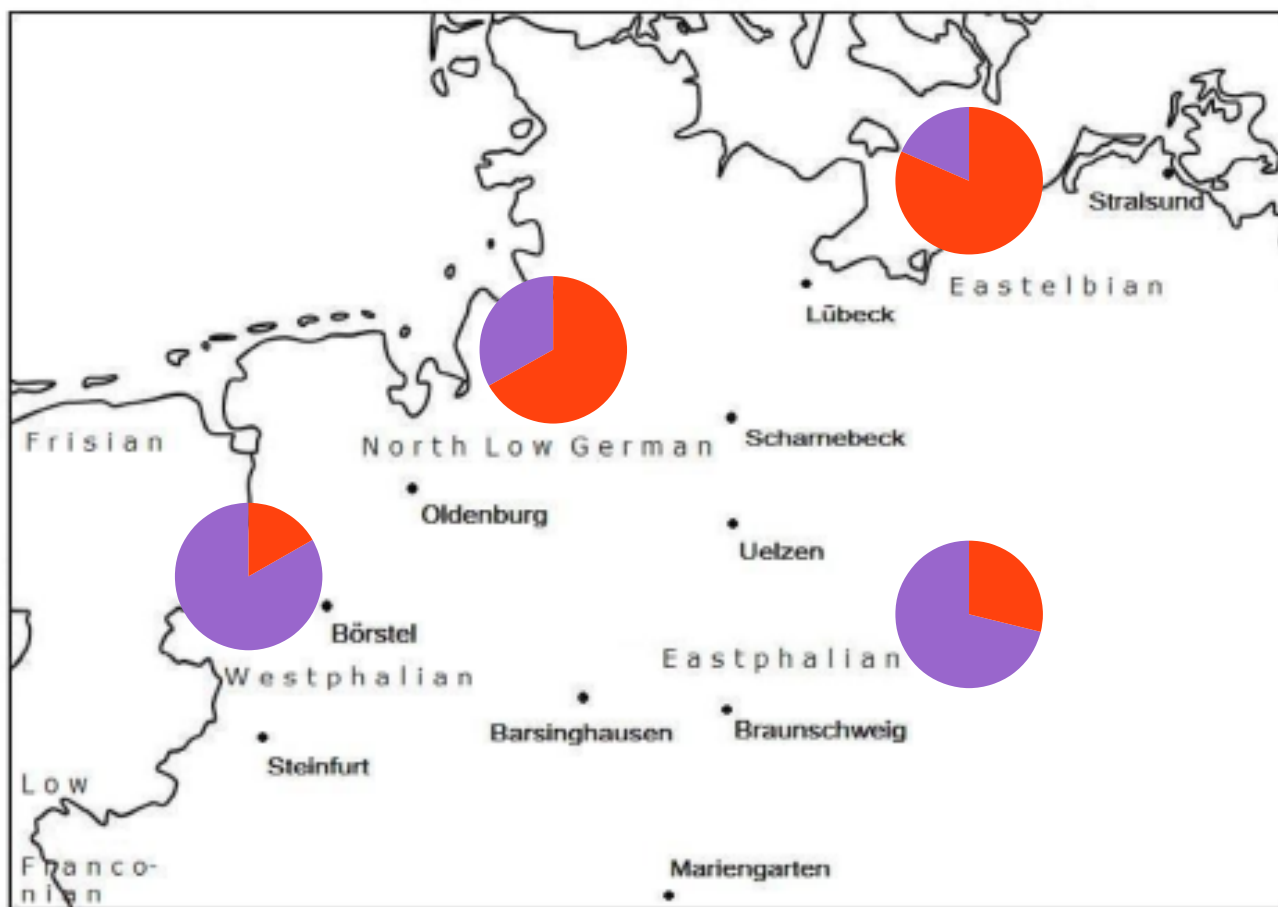
Negation

Null subjects

Case

Conclusion

1375–1424



Introduction

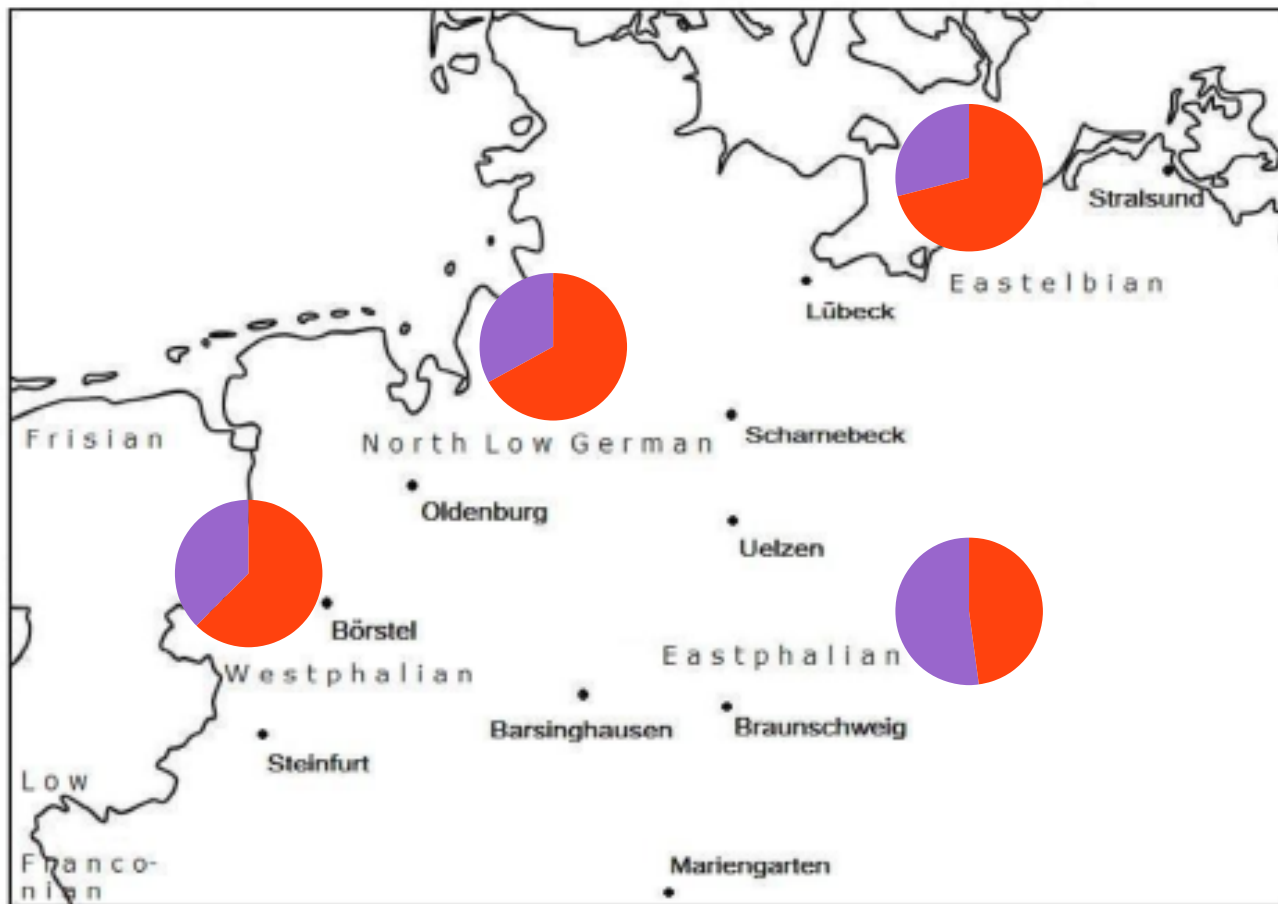
Negation

Null subjects

Case

Conclusion

1425–1474



Introduction

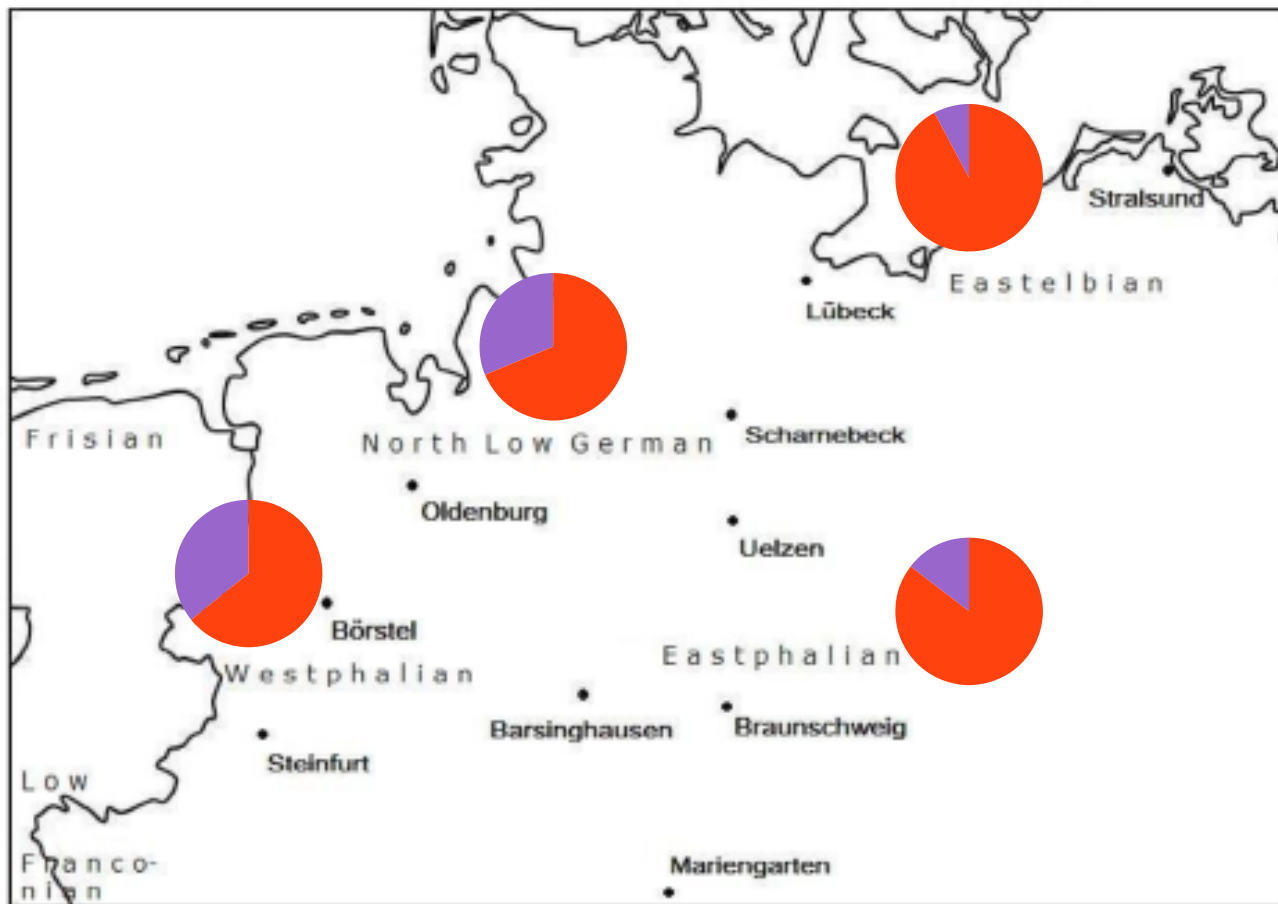
Negation

Null subjects

Case

Conclusion

1475–1524



Introduction

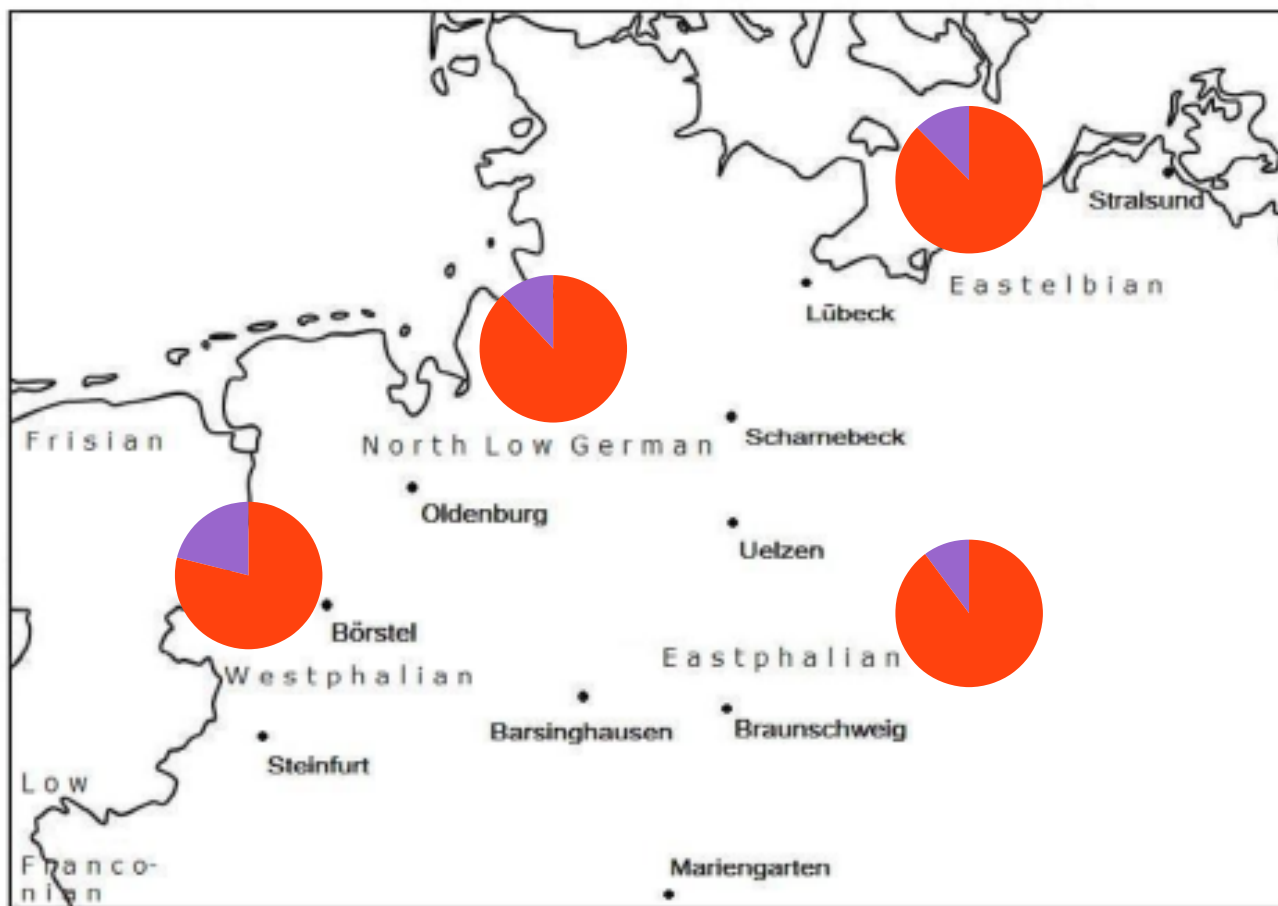
Negation

Null subjects

Case

Conclusion

1525–1574



Introduction

Negation

Null subjects

Case

Conclusion

Sociolinguistic typology

- Preverbal negator *en/ne* (the bearer of the uninterpretable feature) is lost much faster and earlier in the north-eastern Hansa cities (e.g. Lübeck, Stralsund) than in West- and Eastphalian. Why?
 - North-eastern area was colonized later; area formerly inhabited by Slavonic speakers.
 - Cities such as Lübeck and Stralsund were centres of Hanseatic trade.
 - Typical scenario of urbanization/levelling/koinéization (cf. Peters 2000: 1414)
 - Situation of ‘receptive multilingualism’ (Braunmüller 2007).

Sociolinguistic typology

- This is consistent with the prediction made in the introduction.
- Other evidence from Jespersen's Cycle beyond Middle Low German:
 - French
 - Dutch
 - English (?)

Introduction

Negation

Null subjects

Case

Conclusion

French (Montreal)



- In Montreal French, a new world variety, the original negative particle *ne* is almost completely absent (Sankoff & Vincent 1977).
 - The variety arose in a colonization scenario with short-term adult language contact in the 17th c. between speakers of different Oïl languages
 - Paris koiné (itself a levelled output of migrant dialects) used as their lingua franca (Wittman 1995).

Introduction

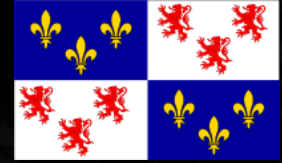
Negation

Null subjects

Case

Conclusion

French (Picard)



- In the Picard dialect of the north of France, *ne* is rarely if ever omitted (Coveney 1996: 62; Auger & Villeneuve 2008).
 - Here, two typologically and historically close varieties are in **long-term** contact (all speakers of Picard are bilingual with French)
 - Scenario of receptive multilingualism (cf. the discussion whether Picard is a dialect of French or a separate language, reported in Auger & Villeneuve 2008) without leading to the loss of *ne*.

Introduction

Negation

Null subjects

Case

Conclusion



In France & Belgium



- Pohl (1968: 1352): *ne* is lost...
 - more in France than in Belgium (?)
 - more in Paris than in the provinces
 - more in towns than in the countryside
 - more among monolinguals than among bilingual/bidialectal speakers (!)
- Pohl and Coveney (1996) argue that socio-economic factors contributed to the loss of *ne*, accelerating in the mid-19th century: specifically, the opening and extension of social networks as a result of the development of the railway and the migration to towns and cities.

Introduction

Negation

Null subjects

Case

Conclusion



Dutch & Flemish



- The old preverbal negation particle *en* was lost much earlier from northern Dutch dialects (during the 17th century) than from southern ones (beginning in the 19th century, still present in some Flemish dialects today; Burridge 1993, Beheydt 1998).
- After their independence at the end of the 80 Years' War, the northern provinces, especially Holland, enjoyed a great economical upturn (Gouden Eeuw 'golden age/century') and became an international centre of trade and intellectual life.
- This attracted a large amount of migration from within the low countries and from outside, resulting in dialect levelling and koinéization (Breitbarth 2013b).

Introduction

Negation

Null subjects

Case

Conclusion

English



- The evidence from Middle English is difficult to interpret due to imbalance in the attested corpus (see Wallage 2008: 649).
- However, Einkenel (1912) shows that Northern varieties retain *ne* longer than Southern and Midlands varieties.
- Contact with Scandinavian? (Though Trudgill 2011 argues that this is not the right type of contact for simplification.)

Introduction

Negation

Null subjects

Case

Conclusion

Negation: summary

- Evidence from the diachrony of negation seems to fit quite well with our hypothesis: the preverbal negator is lost first, and fastest, in situations of intense short-term contact.
- More work needs to be done, particularly beyond Indo-European.
 - e.g. Libyan Arabic, Nafusi Berber (Ghadgoud 2013)

Introduction

Negation

Null subjects

Case

Conclusion

Case study 2

- The loss of null subjects.
 - The development and loss of null subjects has been argued to be a cyclical development much like Jespersen's Cycle (cf. Rowlett 1998: 136, Faarlund 2011, van Gelderen 2011, Walkden 2012):
 - Subject pronouns are grammaticalized as ('rich') agreement markers (Stage I)
 - These markers are reinforced by new subject pronouns (Stage II)
 - Markers are lost; pronouns become obligatory (Stage II)
 - between stages II and III is a partial null subject stage (Walkden 2012)

Introduction

Negation

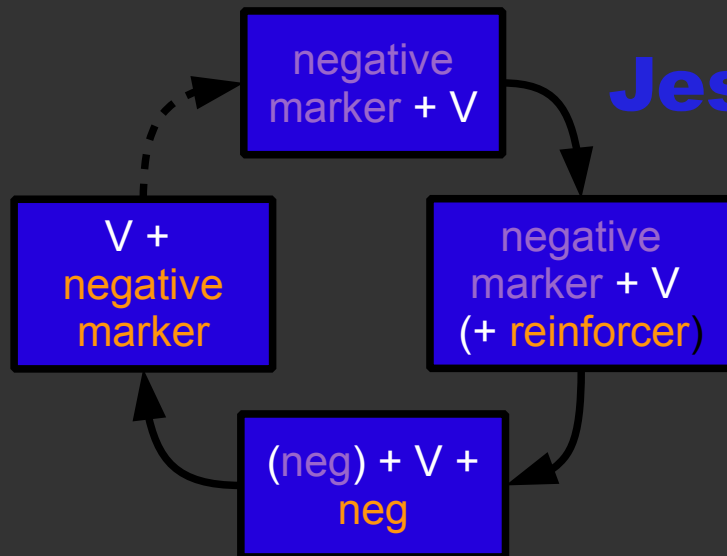
Null subjects

Case

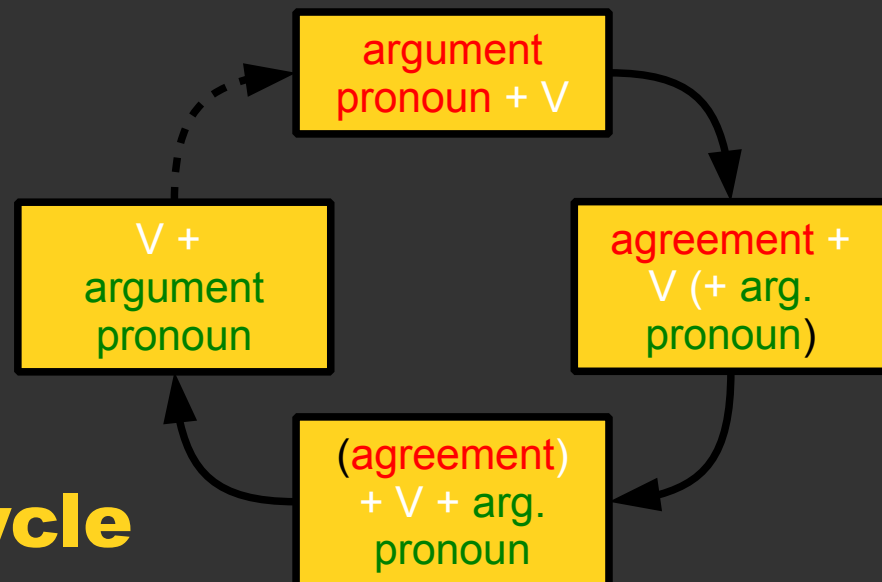
Conclusion

The null argument cycle

Jespersen's Cycle



Null Argument Cycle



Introduction

Negation

Null subjects

Case

Conclusion

Analysis of null subjects

- Consistent null subject languages (NSLs) have definite null subjects, which are typically obligatory in some contexts (e.g. obviation effects), but no generic inclusive null subjects (null *one*)
 - “the property which consistent NSLs have, that partial NSLs don’t have, is a D(efinite)-feature as part of the ϕ -feature make-up of finite T” (Holmberg 2010: 94)
 - Checked/valued by Agree with a null topic in the C-domain (cf. Frascarelli 2007)
 - Allows a ϕ P subject to incorporate into T^0 , as its features are a subset of those of the probe (based on theory of head-movement in Roberts 2010)

Introduction

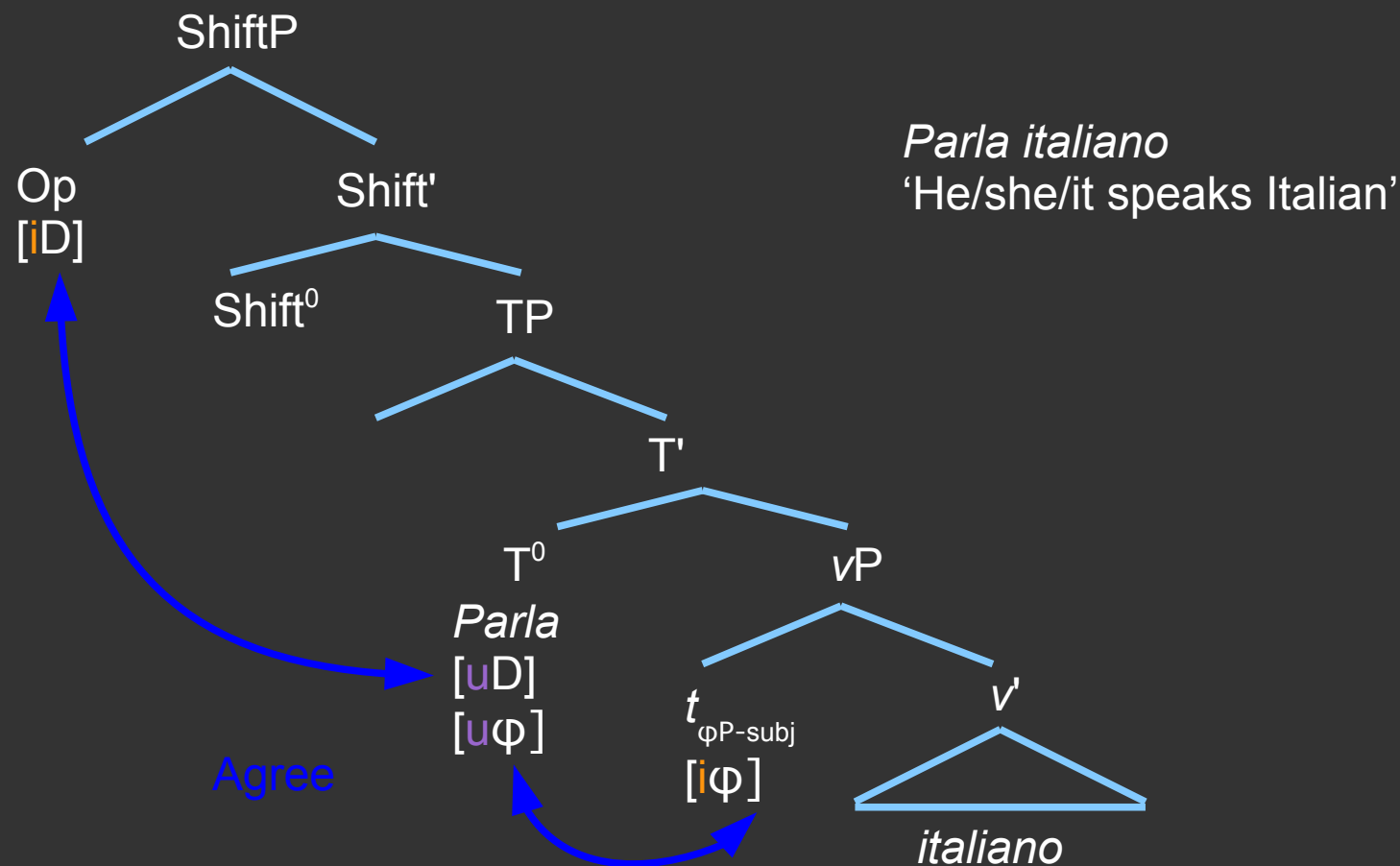
Negation

Null subjects

Case

Conclusion

Analysis of consistent NSLs



Introduction

Negation

Null subjects

Case

Conclusion

Partial NSLs

- Partial NSLs have definite null subjects only in some contexts (e.g. 1st & 2nd person in Finnish), but these are not obligatorily null, and also generic inclusive null subjects (null *one*)
 - No [μ D] feature on T^0 ; instead must have DP subjects that become null by chain reduction

Introduction

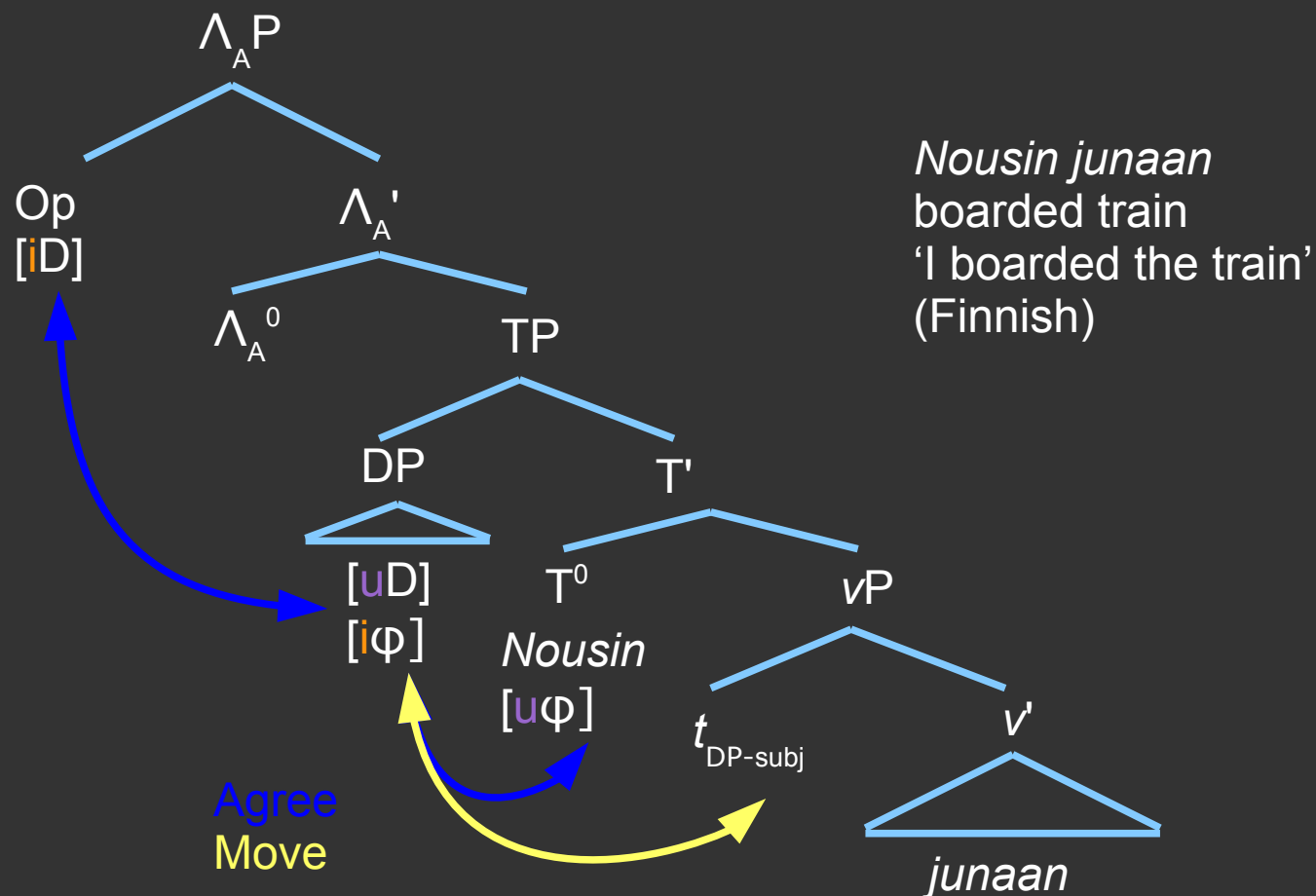
Negation

Null subjects

Case

Conclusion

Analysis of partial NSLs



Introduction

Negation

Null subjects

Case

Conclusion

The loss of null subjects

- Historical record shows many partial NSLs have developed from consistent NSLs:
 - Brazilian Portuguese (partial) < early European Portuguese (full) (Roberts 2011)
 - Marathi (partial) < Sanskrit (full) (Kiparsky 2009)
 - Russian (partial) < $\approx$ Old Church Slavonic (full) (Eckhoff & Meyer 2011)
 - Hebrew (partial) < $\approx$ Biblical Hebrew (full) (Holmstedt 2010)
- Crucial feature in transition from consistent to partial NSL: loss of [uD] feature on T^0 .

Introduction

Negation

Null subjects

Case

Conclusion

The loss of null subjects

- Historical record shows (less convincingly) that some partial NSLs have developed into non-NSLs:
 - Standard Finnish (partial) $\approx$ > Colloquial Finnish (non-NSL) (Holmberg 2010)
 - Old English (partial) > Modern English (non-NSL) (Walkden 2012: ch. 5)
- Crucial feature in transition from partial NSL to non-NSL: loss of [uD] on DPs.

Introduction

Negation

Null subjects

Case

Conclusion

L2 acquisition of null subjects

- L2 acquisition literature shows that null subjects are (in general) not produced at a target-like rate by L2 acquirers:
 - Bini (1993): L1 Spanish speakers learning Italian (both consistent NSLs!) systematically overproduce 'redundant' overt pronouns in their L2
 - Sorace et al. (2009: 464): L2 learners of any null subject language appear to “use overt subject pronouns as a compensatory ‘default’ strategy”, regardless of the structure of their L1

Testing ground 1

- **Heritage languages.** We predict loss of null subject status, *to the extent that L2 acquisition plays a major role.*
 - **Yes:** Otheguy, Zentella & Livert (2007). In heritage varieties of Spanish used in New York, the rate of pronominal subject use is much higher in second- and third-generation speakers than in Spanish-born speakers.
 - **Yes:** Heap & Nagy (1998). In Faetar – a Francoprovençal partial NSL variety spoken in southern Italy – apparent-time data indicate increased use of subject pronouns.
 - **Yes:** Chociej (2011). In heritage Polish spoken in Toronto, the rate of pronominal subject use is much higher in second- and third-generation speakers than in Spanish-born speakers.

Introduction

Negation

Null subjects

Case

Conclusion

Testing ground 1

- **Heritage languages.** We predict loss of null subject status, *to the extent that L2 acquisition plays a major role.*
 - **No:** Torres Cacoullos & Travis (2011). English code-switching in New Mexico Spanish does not seem to increase the use of first-person subject pronouns (once priming effects are factored in).
 - **No:** Nagy et al. (2011). Cantonese, Italian and Russian heritage language subject pronoun use in Toronto not affected by generation, identity etc.

Introduction

Negation

Null subjects

Case

Conclusion

Testing ground 2

- **New world and colonial varieties** of NSLs. These are likely to present a prime case of dialect levelling/koinéization due to the circumstances of settlement.
 - Brazilian Portuguese vs. European Portuguese: same starting point, but clear (and historically visible) shift to partial NSL in the former.
 - Caribbean Spanish vs. General Spanish (Toribio 1996, Kato 2012): again, same starting point, but shift to partial NSL in the former.
 - Chabacano (Philippine Creole Spanish) (Lipski 1999): partial NSL with generic inclusive null subject.

Introduction

Negation

Null subjects

Case

Conclusion

Null subjects: summary

- Certain instances of loss of null subjects seem to support the hypothesis. However, the data (and interpretation) are not always clear:
 - Do heritage languages undergo the right kind of contact (short-term, involving L2 acquisition) for simplification?
 - Since English is a non-NSL, do studies of contact with English (e.g. all of the heritage language studies mentioned above, except Heap & Nagy 1998) really tell us anything?
 - Could there be an interaction with concomitant loss of rich agreement morphology (also predicted under Trudgill's account)?

Introduction

Negation

Null subjects

Case

Conclusion

Case study 3

- **Case.**
 - Bentz & Winter (2013) demonstrate synchronic correlations between:
 - increased numbers of L2 speakers and absence of (overt) nominal case
 - increased numbers of L2 speakers and reduced numbers of cases
 - They argue from this that a preponderance of L2 speakers leads to the loss of case(s).

Case

- If Case features are always uninterpretable, following Chomsky (1995: 278–279), then we predict their loss in short-term L2 acquisition situations.
- In Germanic:
 - overt case is lost in the histories of Dutch, Mainland Scandinavian and English
 - the distinction between accusative and dative is lost in the history of Low German.

Case

- But more work is needed to see whether abstract Case is really involved, and how.
- The claim that abstract Case can be lost isn't (completely) crazy; see Sheehan & van der Wal (2013) for empirical and conceptual arguments that Case might be absent in some languages, e.g. the Bantu language Lusoga.
 - The sociohistorical background of such languages would need to be investigated.

Introduction

Negation

Null subjects

Case

Conclusion

Interlude: complexification

- **Additive** complexification (i.e. transfer) under this approach specifically involves the transfer of **uninterpretable** features from one grammar to another.
- Additive complexification is predicted to arise typically in situations involving long-term stable bilingualism.
 - *Sprachbund* phenomena bear this out.

Introduction

Negation

Null subjects

Case

Conclusion

Interlude: **complexification**

- **Spontaneous** (non-additive) complexification, i.e. the emergence of new **uninterpretable** features, is likely to occur in situations without contact (e.g. Trudgill's 'traditional dialects').
 - The emergence of **uninterpretable** features from **interpretable** features is a reasonably well-understood phenomenon, and is characteristic of grammaticalization (van Gelderen 2009, 2011)

Introduction

Negation

Null subjects

Case

Conclusion

Conclusion: focus on L2

- Traditionally, following Lightfoot (1979) and ultimately Paul (1880), work in diachronic generative syntax has accorded a central role to L1 acquisition in language change.
 - This centrality has been questioned (e.g. Diessel 2012).
 - L1 acquisition can't *not* be involved in language change, but...
 - ...it doesn't really help us approach the actuation problem.

Introduction

Negation

Null subjects

Case

Conclusion

Conclusion: focus on L2

- The approach to syntactic change taken here accords a central position to **L2 acquirers** and their subsequent influence on the PLD of new L1 acquirers.
 - In a nutshell: the type of change a language is likely to undergo, and the time and rate at which it is likely to undergo it, depends on the proportion and distribution of L2 acquirers in the population.
 - Not a new insight, but potentially a fruitful research direction.

Introduction

Negation

Null subjects

Case

Conclusion

Caveats

- The approach depends on:
 - the validity of (something like) the Interpretability Hypothesis as a characterization of L2-difficulty
 - the validity of Trudgill's (2011) insight on how different types of acquisition play out at the population level
 - the validity of the individual syntactic analyses proposed
- All of these are questionable!

Conclusion

- Still, the evidence from the diachrony of negation and of null subjects seems to be at least consistent with the basic hypothesis: that in situations involving high levels of adult L2 acquisition and transmission of 'imperfect' adult L2 varieties, at the population level, uninterpretable features will be lost.

Introduction

Negation

Null subjects

Case

Conclusion

Acknowledgements

- This talk is based on joint work with Anne Breitbarth (Gent), whose input has been invaluable.
- Thanks also to current and former colleagues and friends at Cambridge and Manchester.

Introduction

Negation

Null subjects

Case

Conclusion



Thank you!

References (1)

- Aikhenvald, Alexandra Y. 2003. A grammar of Tariana, from northwest Amazonia. Cambridge: Cambridge University Press.
- Al-Thubaiti, Kholoud. 2009. The acquisition of uninterpretable features: the case of vP ellipsis. In Melissa Bowles, Tania Ionin, Silvina Montrul & Annie Tremblay (eds.), Proceedings of the 10th Generative Approaches to Second Language Acquisition Conference (GASLA 2009), 192–200. Somerville, MA: Cascadia Proceedings Project.
- Auger, Julie, & Anne-José Villeneuve. 2008. Ne deletion in Picard and in regional French: evidence for distinct grammars. In Miriam Meyerhoff & Naomi Nagy (eds.), Social lives in language: sociolinguistics and multilingual speech communities celebrating the work of Gillian Sankoff, 223–247. Amsterdam: John Benjamins.
- Baker, Mark. 2008. The macroparameter in a microparametric world. In Theresa Biberauer (ed.), The limits of syntactic variation, 351–373. Amsterdam: John Benjamins.
- Beheydt, Griet. 1998. Het gebruik en de vorm van de negatie in het zuidelijke Nederlands in een diachronisch perspectief (15e–20e eeuw). Licentiate's thesis. University of Leuven.

Introduction

Negation

Null subjects

Case

Conclusion

References (2)

- Bentz, Christian, & Bodo Winter. 2013. Languages with more second-language learners tend to lose nominal case. *Language Dynamics and Change* 3, 1–27.
- Bini, Milena. 1993. La adquisición del italiano: más allá de las propiedades sintácticas del parámetro pro-drop. In Juana M. Liceras (ed.), *La lingüística y el análisis de los sistemas no nativos*, 126–139. Ottawa: Dovehouse.
- Borer, Hagit. 1984. *Parametric syntax*. Dordrecht: Foris.
- Braunmüller, Kurt. 2007. Receptive multilingualism in Northern Europe in the Middle Ages: a description of a scenario. In Jan ten Thije & Ludger Zeevaert (eds.), *Receptive multilingualism and intercultural communication*, 25–47. Amsterdam: John Benjamins.
- Breitbarth, Anne. 2008. The development of negation in Middle Low German. Paper presented at the Annual Meeting of the LAGB 2008, University of Essex.
- Breitbarth, Anne. 2013a. Dialect contact and the speed of Jespersen's cycle in Middle Low German. Ms., Universiteit Gent.

Introduction

Negation

Null subjects

Case

Conclusion

References (3)

- Breitbarth, Anne. 2013b. Negation in the history of Low German and Dutch. In Anne Breitbarth, Christopher Lucas and David Willis (eds), *The development of negation in the languages of Europe and the Mediterranean*, vol. 1: case studies, 190–238. Oxford: Oxford University Press.
- Burridge, Kate. 1993. *Syntactic change in Germanic: aspects of language change in Germanic with particular reference to Middle Dutch*. Amsterdam: John Benjamins.
- Campos Dintrans, Gonzalo Santiago. 2011. *Acquisition of morphosyntax in the adult second language: the phonology factor*. MA dissertation. University of Iowa.
- Chociej, Joanna. 2011. *Polish null subjects: English influence on heritage Polish in Toronto*. Ms., University of Toronto.
- Chomsky, Noam. 1963. Formal properties of grammar. In R. D. Luce, R. R. Bush and E. Galanter (eds.), *Handbook of Mathematical Psychology*, vol. 2, 323–418. New York: Wiley.
- Chomsky, Noam. 1995. *The Minimalist Program*. Cambridge, MA: MIT Press.
- Chomsky, Noam. 2001. Derivation by phase. In Michael Kenstowicz (ed.), *Ken Hale: a life in language*, 1–53. Cambridge, MA: MIT Press.

Introduction

Negation

Null subjects

Case

Conclusion

References (4)

- Coveney, Aidan. 1996. Variability in spoken French: a sociolinguistic study of interrogation and negation. Exeter: Elm Bank.
- Dahl, Östen. 2004. The growth and maintenance of linguistic complexity. Amsterdam: John Benjamins.
- Eckhoff, Hanne, & Roland Meyer. 2011. Conditions on null subjects in Old Church Slavonic: a contrastive view. Paper presented at the Workshop on the Diachrony of Referential Null Arguments, 20th International Conference on Historical Linguistics, Osaka.
- Einkenkel, Eugen. 1912. Die englische Verbalnegation, ihre Entwicklung, ihre Gesetze und ihre zeitlich-örtliche Verwendung. Anglia 35, 187–248.
- Faarlund, Jan Terje. 2011. The pro cycle. Paper presented at ICHL 20, Osaka, July 2011.
- Farrar, Kimberley, & Mari C. Jones. 2002. Introduction. In Mari C. Jones and Edith Esch (eds.), Language change: the interplay of internal, external, and extra-linguistic factors, 1–16. Berlin: Mouton de Gruyter.

Introduction

Negation

Null subjects

Case

Conclusion

References (5)

- Frascarelli, Mara. 2007. Subjects, topics and the interpretation of referential pro: an interface approach to the linking of (null) pronouns. *Natural Language and Linguistic Theory* 25, 691–734.
- Ghadgoud, Khawla M. 2013. Jespersen's Cycle in Libyan Arabic and Nafusi Berber. MA dissertation, University of Manchester.
- Hale, Mark. 1998. Diachronic syntax. *Syntax* 1, 1–18.
- Hawkins, John. 1994. A performance theory of order and constituency. Cambridge: Cambridge University Press.
- Hawkins, John. 2004. Efficiency and complexity in grammars. Oxford: Oxford University Press.
- Hawkins, Roger, & Hajime Hattori. 2006. Interpretation of English multiple wh-questions by Japanese speakers: a missing uninterpretable feature account. *Second Language Research* 22, 269–301.
- Heap, David, & Naomi G. Nagy. 1998. Subject pronoun variation in Faetar and Francoprovençal. *Papers in Sociolinguistics. N.WAVE-26 à l'Université Laval*, 291–300. Québec: Nota bene.

Introduction

Negation

Null subjects

Case

Conclusion

References (6)

- Hockett, Charles F. 1958. A course in modern linguistics. New York: Macmillan.
- Holmberg, Anders. 2010. Null subject parameters. In Theresa Biberauer, Anders Holmberg, Ian Roberts & Michelle Sheehan (eds.), Parametric variation: null subjects in Minimalist theory, 88–124. Cambridge: Cambridge University Press.
- Holmstedt, Robert. 2010. Pro-drop in Hebrew: a summary.
<http://ancienthebrewgrammar.wordpress.com/2010/06/05/pro-drop-in-hebrew/>.
Last accessed: 9th May 2013.
- Ihalainen, Ossi. 1991. On grammatical diffusion in Somerset folk speech. In Peter Trudgill & J. K. Chambers (eds.), Dialects of English: studies in grammatical variation, 148–160. London: Longman.
- Kato, Mary A. 2012. Brazilian Portuguese and Caribbean Spanish: similar changes in Romania Nova. In Charlotte Galves, Sonia Cyrino, Ruth Lopes, Filomena Sandalo & Juanito Avelar (eds.), Parameter theory and linguistic change, 117–132. Oxford: Oxford University Press.
- Kiparsky, Paul. 2009. On the architecture of Pāṇini's grammar. In Gérard Huet, Amba Kulkarni & Peter Scharf (eds.), Sanskrit computational linguistics, 33–94. Berlin: Springer.

Introduction

Negation

Null subjects

Case

Conclusion

References (7)

- Lardiere, Donna. 2005. On morphological competence. In Laurent Dekydtspotter, Rex A. Sprouse & Audrey Liljestrang (eds.), *Proceedings of the 7th Generative Approaches to Second Language Acquisition Conference (GASLA 2004)*, 178–192. Somerville, MA: Cascadilla Proceedings Project.
- Lardiere, Donna. 2008. Feature assembly in second language acquisition. In Juana Liceras, Helmut Zobl & Helen Goodluck (eds.), *The role of features in second language acquisition*, 106–140. Mahwah, NJ: Erlbaum.
- Lardiere, Donna. 2009. Some thoughts on a contrastive analysis of features in second language acquisition. *Second Language Research* 25, 173–227.
- Lenneberg, Eric. 1967. *Biological foundations of language*. New York: Wiley.
- Lightfoot, David. 1979. *Principles of diachronic syntax*. Cambridge: Cambridge University Press.
- Lipski, John M. 1999. Null subjects in (Romance-derived) creoles: routes of evolution. Paper presented at the Annual Meeting of the Society for Pidgin and Creole Linguistics, Los Angeles, January 1999.

Introduction

Negation

Null subjects

Case

Conclusion

References (8)

- Mobbs, Iain. 2008. 'Functionalism', the design of the language faculty, and (disharmonic) typology. MPhil dissertation. University of Cambridge.
- Nagy, Naomi G., Nina Aghdasi, Derek Denis, & Alexandra Motut. 2011. Null subjects in heritage languages: contact effects in a cross-linguistic context. University of Pennsylvania Working Papers in Linguistics 17.
<http://repository.upenn.edu/pwpl/vol17/iss2/16>. Last accessed: 9th May 2013.
- Otheguy, Ricardo, Ana Celia Zentella & David Livert. 2007. Language and dialect contact in Spanish in New York: Towards the formation of a speech community. *Language* 83, 1–33.
- Owens, Jonathan. 1997. Arabic-based pidgins and creoles. In Sarah G. Thomason (ed.), *Contact languages: a wider perspective*, 125–172. Amsterdam: John Benjamins.
- Owens, Jonathan. 2001. Creole Arabic: the orphan of all orphans. *Anthropological Linguistics* 43, 348–378.
- Paul, Hermann (1880). *Prinzipien der Sprachgeschichte*. 1st edn. Halle: Max Niemeyer.

Introduction

Negation

Null subjects

Case

Conclusion

References (9)

- Peters, Robert. 2000. Soziokulturelle Voraussetzungen und Sprachraum des Mittelniederdeutschen. In Werner Besch, Anne Betten, Oskar Reichmann & Stefan Sonderegger (eds.), *Sprachgeschichte: Ein Handbuch zur Geschichte der deutschen Sprache und ihrer Erforschung*, 1408–1422. Berlin: Mouton de Gruyter.
- Pohl, Jacques. 1968. Ne dans le français parlé contemporain: les modalités de son abandon. In Antonio Quilis Morales, Ramón B. Carril & Margarita Cantarero (eds.), *Actas del XI Congreso internacional de lingüística y filología románicas*, 1343–1359. Madrid: Revista de Filología Española.
- Roberts, Ian. 2010. Agreement and head movement: clitics, incorporation and defective goals. Cambridge, MA: MIT Press.
- Roberts, Ian. 2011. Taraldsen's Generalization and diachronic syntax: two ways to lose null subjects. In Peter Svenonius (ed.), *Festschrift for Tarald Taraldsen*. Oxford: Oxford University Press.
- Roberts, Ian, & Anna Roussou. 2003. *Syntactic change: a Minimalist approach to grammaticalization*. Cambridge: Cambridge University Press.
- Rowlett, Paul. 1998. *Sentential negation in French*. Oxford: Oxford University Press.

Introduction

Negation

Null subjects

Case

Conclusion

References (10)

- Schwartz, Bonnie D., & Rex A. Sprouse. 1994. Word order and nominative case in non-native language acquisition: a longitudinal study of (L1 Turkish) German interlanguage. In Teun Hoekstra & Bonnie D. Schwartz (eds.), *Language acquisition studies in generative grammar*, 317–368. Amsterdam: John Benjamins.
- Schwartz, Bonnie D., & Rex A. Sprouse. 1996. L2 cognitive states and the Full Transfer/Full Access Model. *Second Language Research* 12, 40–72.
- Sheehan, Michelle, & Jenneke van der Wal. 2013. The worst Case scenario. Paper presented at SyntaxLab, Cambridge, September 2013.
- Slabakova, Roumyana. 2006. Is there a critical period for the acquisition of semantics? *Second Language Research* 22, 302–338.
- Slabakova, Roumyana. 2008. *Meaning in the Second Language*. Berlin: Mouton de Gruyter.
- Slabakova, Roumyana. 2009. What is easy and what is hard to acquire in a second language? In Melissa Bowles, Tania Ionin, Silvina Montrul & Annie Tremblay (eds.), *Proceedings of the 10th Generative Approaches to Second Language Acquisition Conference (GASLA 2009)*, 280–294. Somerville, MA: Cascadilla Proceedings Project.

Introduction

Negation

Null subjects

Case

Conclusion

References (11)

- Sorace, Antonella. 2000. Syntactic optionality in non-native grammars. *Second Language Research* 16, 93–102.
- Sorace, Antonella. 2003. Near-nativeness. In Catherine J. Doughty & Michael H. Long (eds.), *The handbook of second language acquisition*, 130–151. Oxford: Blackwell.
- Sorace, Antonella. 2011. Pinning down the concept of “interface” in bilingualism. *Linguistic Approaches to Bilingualism* 1, 1–33.
- Sorace, Antonella, Ludovica Serratrice, Francesca Filiaci & Michela Baldo. 2009. Discourse conditions on subject pronoun realization: testing the linguistic intuitions of older bilingual children. *Lingua* 119, 460–477.
- Szmrecsányi, Benedikt. 2004. On operationalizing syntactic complexity. In Gérard Purnelle, Cédric Fairon & Anne Dister (eds.), *Le poids des mots: proceedings of the 7th International Conference on Textual Data Statistical Analysis*, vol. 2, 1032–1039. Louvain-la-Neuve: Presses universitaires de Louvain.
- Toribio, Almeida J. 1996. Dialectal variation in the licensing of null referential and expletive subjects. In Carlos Prodi, Carlos Quicoli, Mario Saltarelli & Maria Luisa Zubizarreta (eds.), *Aspects of Romance linguistics*, 409–432. Washington, DC: Georgetown University Press.

Introduction

Negation

Null subjects

Case

Conclusion

References (12)

- Torres Cacoullos, Rena, & Catherine E. Travis. 2011. Testing convergence via code-switching: priming and the structure of variable subject expression. *International Journal of Bilingualism* 15, 241–267.
- Trotzke, Andreas, & Jan-Wouter Zwart. 2013. The complexity of Narrow Syntax. In Frederick J. Newmeyer & Laurel B. Preston (eds.), *Measuring linguistic complexity*. Oxford: Oxford University Press. To appear.
- Trudgill, Peter. 2011. *Sociolinguistic typology: social determinants of linguistic complexity*. Oxford: Oxford University Press.
- Tsimpli, Ianthi Maria, & Maria Dimitrakopoulou. 2007. The Interpretability Hypothesis: evidence from wh-interrogatives in second language acquisition. *Second Language Research* 23, 215–242.
- van Gelderen, Elly. 2009. Renewal in the left periphery: economy and the complementiser layer. *Transactions of the Philological Society* 107, 131–195.
- van Gelderen, Elly. 2011. *The linguistic cycle: language change and the language faculty*. Oxford: Oxford University Press.

Introduction

Negation

Null subjects

Case

Conclusion

References (13)

- Walkden, George. 2012. Syntactic reconstruction and Proto-Germanic. PhD dissertation. University of Cambridge.
- Wallage, Phillip. 2005. Negation in early English: parametric variation and grammatical competition. PhD dissertation. University of York.
- Wallage, Phillip. 2008. Jespersen's Cycle in Middle English: parametric variation and grammatical competition. *Lingua* 118, 643–674.
- Wittmann, Henri. 1995. Grammaire comparée des variétés coloniales du français populaire de Paris du 17e siècle et origines du français québécois. In Robert Fournier & Henri Wittmann (eds.), *Le français des Amériques*, 281–334. Trois-Rivières: Presses universitaires de Trois-Rivières.
- Zeijlstra, Hedde. 2004. Sentential negation and negative concord. PhD dissertation, University of Amsterdam.

Introduction

Negation

Null subjects

Case

Conclusion